



Team: *Motor Group*

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1. Progress report

Achieved since the Introduction Lecture :

- Agreed on a standard testbench design
- Designed schematics for our setup
- Collected some basic Microphone and IMU test data
- Agreed on basic and advanced data labels
- Ordered all the necessary components
- Started work on signal preprocessing
- Finished a basic Micropython script

Activities planned until next meeting:

- Record a sufficient amount of data for our models
- Design 3 AI models: IMU, Microphone, IMU + Microphone

Issues / Support needed

- IMU insensitivity
- Weird storage behaviour
- Minor Deepcraft performance issues
- Question: Training a model on two data streams from different sensors



2. Progress report

Achieved since last meeting:

- Collected a lot of data for both audio and IMU based models
- Started developing and optimizing both microphone and IMU models
- Tweaked the Micropython code to fit our current testbench setup
- Created script for improved impact data collection (IMU for now)

Activities planned until next meeting

- Optimize further for model accuracy
- Create a script for optimized impact training for the microphone
- Convert and upload the models to the board
- Combine the microphone and IMU data into a singular model

Issues / Support needed

- Why is label overlapping not allowed?
- IMU and Mic data not in sync when recording both simultaneously
- Model fails to train
- Filtering training data before model training – necessary or not?

Proposed features for Deepcraft studio:

- Add a confirmation window before recording over unsaved data
- “magnetic” snapping of labels next to each other to prevent overlap
- Feature to fill label gaps with other labels
- Add the predefined label tabs back to an already saved file
- Trimming of recorded data built in
- Continue an already existing recording
- Verbose errors saying at what time the labels are overlapping/outside data



3. Progress report

Achieved since last meeting:

- Finalized and successfully converted the IMU model
- Started work on IMU model deployment
- Started augmenting audio data for the microphone model
- Tested functionality of earlier versions of the model directly on board

Activities planned until next meeting

- Finalize the audio model
- Deploy both models on the board
- Prepare an engaging presentation

Show what you have planned for the project presentation and your submission, you have more room on the next slide

Issues / Support needed

- Model problems
- Lack of training credits
- Email address for Lukas
- Question about how formal/serious the presentation needs to be
- Presentation form: Powerpoint + short demonstration sufficient?

Infineon PSOC6 Lecture presentation and final submission plans

Short project presentation:

Details about the audio model and its development

Details about the IMU model and its development

Short overview of Micropython code

Practical use in the real world?

Any special equipment need for demo at the demo Day ?

We have prepared everything and all we need is your attention :)